

The Renal Biopsy

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KEY POINTS

- Kidney biopsy plays an important role in the management of kidney disease.
- Percutaneous kidney biopsy is generally safe if care is taken to select and prepare the patients beforehand
- Full assessment of the biopsy requires examination by light microscopy, immunohistochemistry and (in most cases) electron microscopy
- The biopsy report should include a morphological description of the biopsy and an interpretation of the appearances in the light of the clinical presentation.
- Recommendations for the essential elements that should be present in the report have been published by the Renal Pathology Society.

INTRODUCTION

The renal biopsy has become a fundamental component in the management of renal disease. Prior to its routine use, only autopsy material was available to investigate the pathophysiology of kidney disease, limiting antemortem diagnosis. However, its development and refinement since the late 1950s have been fundamental for the diagnosis and definition of clinical syndromes and the discovery of new pathological entities.¹ Through the critical analysis of renal biopsies taken at different disease time points, key pathophysiological features of kidney disease have been discovered, which have in turn helped to establish new paradigms in nephrology, and have led to considerable alterations in patient management, with estimates of up to 74% of patients' management changing based on the biopsy results,² and in two-thirds of patients, the biopsy revealed unsuspected diagnostic information. This is true for both native renal biopsies and renal transplant biopsies.³ In addition, much is still being learnt regarding

disease pathogenesis through the study of renal biopsy material, which not only remains a "gold standard" for disease diagnosis, but also has allowed the development of novel biopsy markers, which have revolutionized our concepts of pathological mechanisms.

The first percutaneous kidney biopsies were performed over 50 years ago using a liver biopsy needle and intravenous pyelograms for screening, with the patient either sitting or supine. Their success in obtaining renal tissue and in aiding management confirmed the benefit of the procedure.¹ Many innovations, including using real-time ultrasound, which allows visualization of the needle entering the kidney, spring-loaded needles,⁴ or needle holders, and careful preoperative evaluation of the patient have improved the rate of obtaining renal tissue while minimizing the risks of the procedure.⁵ Consequently, this has placed percutaneous renal biopsy at the very center of modern clinical nephrology. The range of diagnoses for a group of 2219 native kidney biopsies performed at the authors' institution over a 5-year period is shown in Fig. 26.1.

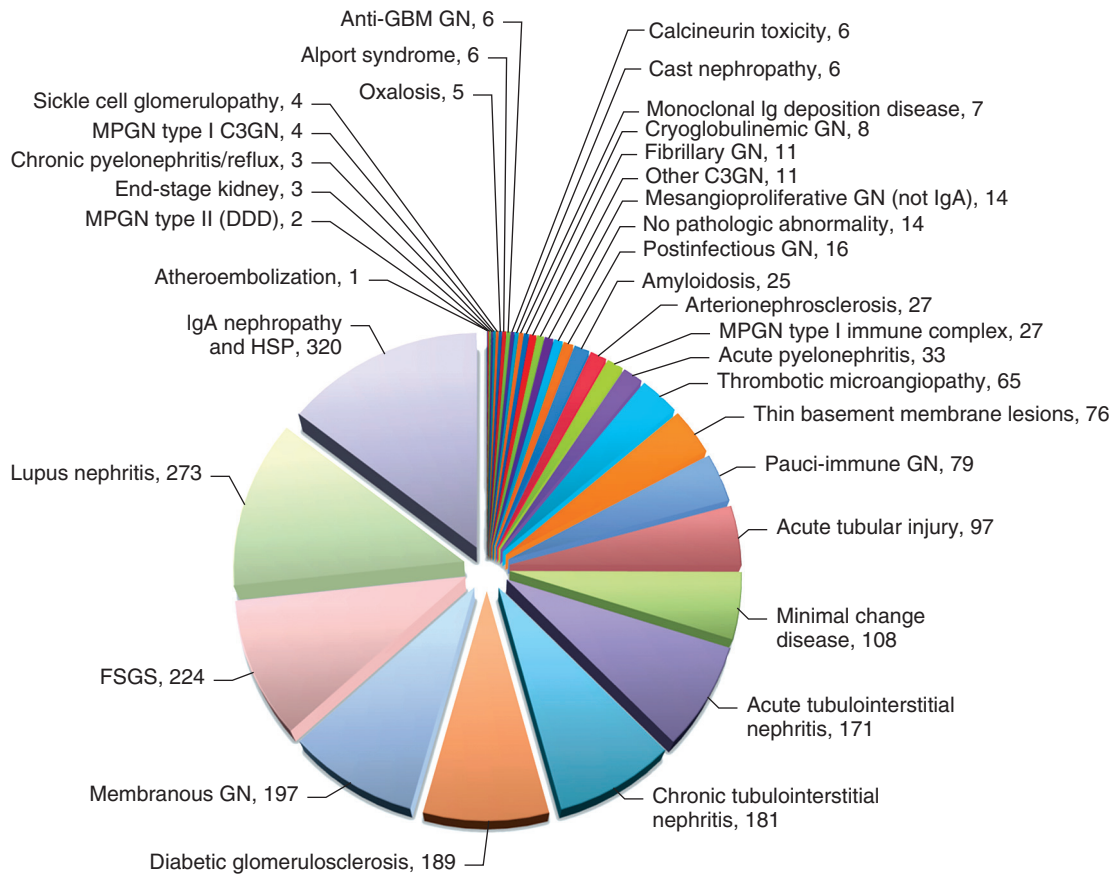


Fig. 26.1 Proportion of diagnoses in 2219 native renal biopsies performed at Hammersmith Hospital over a 5-year period. *DDD*, Dense deposit disease; *FSGS*, focal segmental glomerulosclerosis; *GBM*, glomerular basement membrane; *GN*, glomerulonephritis; *HSP*, Henoch-Schönlein purpura; *Ig*, immunoglobulin; *MPGN*, membranoproliferative glomerulonephritis.

Box 26.1 Indications for Renal Biopsy

- Significant proteinuria (>1 g/day or protein-to-creatinine ratio >100 mg/mmol)
- Microscopic hematuria with any degree of proteinuria
- Unexplained renal impairment (native or transplant kidney)
- Renal manifestations of systemic disease

SAFETY AND COMPLICATIONS OF BIOPSIES

Although generally considered safe, there is a morbidity and a small, but measurable, mortality associated with the procedure, and it is therefore imperative to subject only those patients in whom there will be a potential benefit to these risks. Indications for renal biopsy may vary from one center to another, but accepted indications are listed in [Box 26.1](#). The significant complications related to the procedure are hemorrhage, development of arteriovenous fistulas, and to a lesser extent sepsis.⁶⁻⁸ Bleeding with macroscopic hematuria and the development of perinephric hematomas may be minor and self-resolving or major and require intervention in the form of blood transfusions, embolization, or rarely

surgery. Secondly, there is a risk of formation of arteriovenous fistulas, which may be asymptomatic and spontaneously resolve or lead to a significant vascular steal syndrome, compromising the rest of the kidney through ischemia. Finally, there is the risk of sepsis following the procedure, through the introduction of a septic focus or its dissemination. Overall, the risks of complication vary from center to center and between practitioners, but can be estimated between 3.5% and 13%, with the majority being minor complications (approximately 3%–9%),⁶⁻⁸ although this appears to have decreased in recent years.^{9,10} Mortality from the procedure is generally a result of undiagnosed bleeding with significant hematoma formation and was reported in up to 0.2% of cases from some of the larger biopsy series,^{7,8} although other studies suggest that it represents an extremely rare adverse event.⁵ Some degree of bleeding is common, as approximately one-half of patients have a drop in hemoglobin postbiopsy, and one-third will develop some hematoma, but only in a minority (≤7%) will bleeding be significant and require intervention.^{5,8,11} Complications appear to be more common in native than transplant kidneys, and in patients with more advanced renal impairment, with prolonged bleeding times, or with lower hemoglobin (11 ± 2 vs. 12 ± 2 g/dL).^{7,8} One prospective study identified the only risk factors for bleeding complications as being female, younger patients (35 ± 14.5 years vs. 40.3 ± 15.4 years), and those with a prolonged partial thromboplastin

time.¹² Interestingly, needle size, number of passes, blood pressure, and renal impairment were not different between those with bleeding complications and those without. However, in this study all patients with prolonged bleeding times received DDAVP (1-deamino-8-D-arginine vasopressin) to correct the abnormality and 75% of patients had serum creatinine of less than 132 $\mu\text{mol/L}$. Conversely, others using retrospective univariate analysis have reported that blood pressure of 160/100 mm Hg or higher or a serum creatinine of greater than 2 mg/dL more than doubled the risk of bleeding.^{8,13} Overall, however, no effective means has been established to identify those individuals at risk of developing “clinically” significant complications. In one small series, postbiopsy ultrasound within an hour had a 95% negative predictive value for predicting clinically significant hemorrhagic complications,¹¹ meaning that the absence of a hematoma on the postbiopsy scan was very suggestive of an uncomplicated clinical course. Debate continues regarding the routine use of DDAVP to counteract uremic bleeding tendencies. In part this is because its use was previously reserved for only those patients with prolonged bleeding times, and numerous studies have since demonstrated that complication rates are no different if bleeding time estimation is omitted from the preoperative assessment,^{14,15} as it does not predict clinical complications.¹² However, more recent data from a randomized double blinded trial suggested a significant benefit in preventing bleeding complications with few adverse events.¹⁶ A total of 162 low-risk adult patients undergoing biopsy were enrolled and randomized to subcutaneous DDAVP (0.3 $\mu\text{g/kg}$) or placebo. The patients were normotensive, had preserved renal function with serum creatinine of less than 1.5 mg/dL (estimated glomerular filtration rate >60 mL/min), and demonstrated a significant reduction in postbiopsy bleeds from 30.5% to 13.7% (relative risk 0.45), a significant reduction in hematoma size in those who did bleed, and a reduction in duration of hospital stay. However, hemoglobin drop after biopsy was minimal and there were no major complications, leading some to question the benefit of reduction in clinically unimportant hematomas, which can be frequently found following biopsy if looked for. No thrombotic, hyponatremic, or cardiovascular events were recorded. Whether these data, in patients with preserved renal function, could be translated to those higher-risk patients with greater renal impairment is unclear and is a question worthy of a randomized trial.

Many centers stop antiplatelet therapy prebiopsy for elective procedures, but recent data suggest that bleeding rates may be no different in those taking aspirin or stopping a week beforehand, and may avoid the increased risk of cardiovascular events following aspirin withdrawal.⁹ In a center that does not routinely stop aspirin (but does stop clopidogrel), a retrospective analysis of 2563 biopsies revealed a major bleeding complication in only 2.2%, and in those in whom a complete drug record was available, no significant difference was noted on or off aspirin (357 vs. 1509, respectively, $p = .93$).¹⁰ Very limited data are available on biopsies performed on patients taking clopidogrel.¹⁷

Guidelines on consenting patients and providing appropriate risk estimates have been produced by certain national renal groups and one such example is provided in Table 26.1. These estimates may err on the conservative side and should be adapted to local practice if adequate complication

Table 26.1 Quoted Risks of Renal Biopsy^a

Complication	Quoted Risk
Macroscopic hematuria	1:10
Bleeding that requires a blood transfusion	<1:50
Bleeding that may require urgent x-ray tests or even an operation to stop the bleeding	<1:1500
Severe bleeding necessitating nephrectomy	<1:3000
Deaths	Extremely rare

^aAccording to UK Renal Association (<http://www.renal.org/information-resources/procedures-for-patients>).

Table 26.2 Contraindications to Renal Biopsy

Absolute Contraindications	Relative Contraindications
Uncontrolled hypertension	Single kidney
Bleeding diathesis	Antiplatelet/clotting agents ^a
Widespread cystic disease	Anatomical abnormalities
Hydronephrosis	Small kidneys
Uncooperative patient	Active urinary/skin sepsis
	Obesity ^a

^aAspirin and body mass index >40 kg/m² were not found in recent limited cohort data to be a significant risk.

data are available. As well as developing procedure-related complications, there is the chance that an inadequate core of tissue is obtained for diagnosis, containing too few glomeruli or insufficient cortical material, and this is reported in between 1% and 5% of cases. The size requirements for accurate diagnosis are discussed later.

There are certain absolute contraindications, which preclude percutaneous biopsy, whereas there are a number of relative contraindications (Table 26.2) that may be circumvented depending on the importance of the biopsy, the operator's experience, and the supportive facilities available. Ideally, all efforts should be made to deal with the relative contraindications; however, in the context of acute renal failure this may not always be possible. With modern techniques evidence is emerging that previously perceived high-risk factors such as obesity, plasma cell dyscrasias such as myeloma, or amyloidosis are not actually associated with higher rates of bleeding complications.^{10,18,19} The critical preoperative steps are to ensure that blood pressure is controlled, that the patient does not have a bleeding diathesis or a urinary tract infection, and that the kidneys are suitably imaged, with no evidence of obstruction, widespread cystic disease, or malignancy (although percutaneous biopsy is increasingly used to diagnose the nature of renal masses). As a result, preoperative assessment should allow those patients unsuitable for percutaneous biopsy to be referred for an alternative approach (Fig. 26.2). In these patients, there are other means of obtaining renal tissue, which include open biopsies,²⁰ laparoscopic biopsies, or transjugular biopsies.²¹ Each is associated with certain complications and has particular merits depending on the clinical scenario (Table 26.3). Overall,

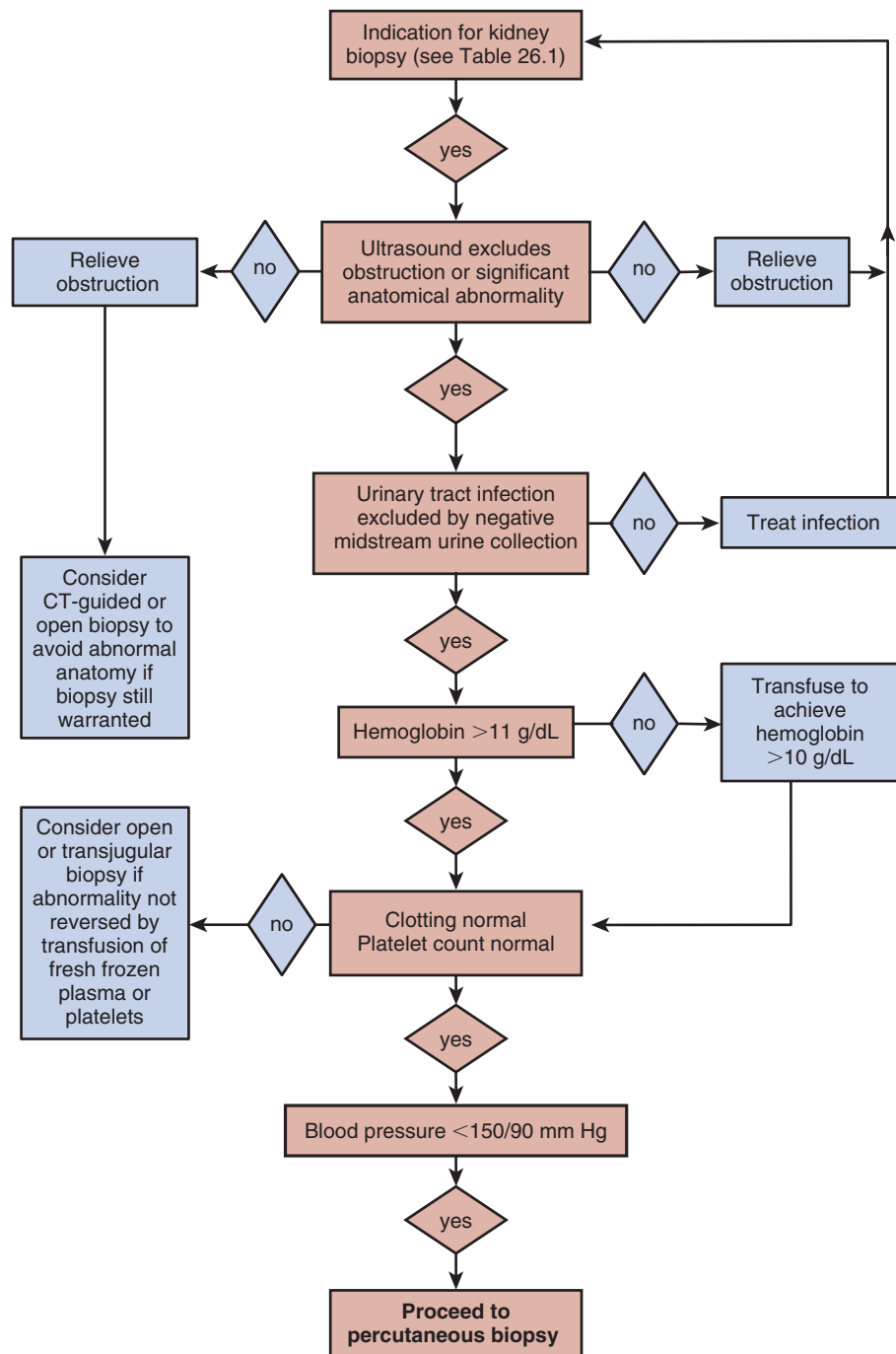


Fig. 26.2 Renal biopsy flowchart. CT, Computed tomography.

these are generally only required for a minority of potential biopsy patients.

The safe duration for observation following renal biopsy has been investigated in a number of studies, which suggest that early discharge (after only 4-hour observation) will result in a number of missed complications, with many more occurring between 8 and 24 hours postprocedure. Even after 8 hours, 23% to 33% of complications will be missed. However, an overnight stay will allow an extra 20% of complications to be identified prior to discharge with between 85% and 95% of complications being identified at 12 hours and 89% and 98% following 24-hour observation.^{7,22} Some units practice a

policy of day biopsies with a minimum 6-hour bed rest period, which is extended only if there is evidence of bleeding, and this appears to be associated with no increased complication rates.²³ Vigilant observation of blood pressure, pulse rate, and evidence of hematuria is required in all cases.

BIOPSY HANDLING

Detailed descriptions of methods of handling biopsies can be found in a number of publications including Churg and colleagues,²⁴ Furness,²⁵ and Walker and colleagues.²⁶

Table 26.3 Alternative Methods for Obtaining Renal Tissue and Their Risks and Benefits Compared With a Percutaneous Approach

Method	Advantage	Disadvantage
Transjugular approach	Can be of use in those with a bleeding diathesis, ventilated patients, or if combined liver and renal biopsy is required	Risk of capsular perforation Inadequate material in up to 24%
Open approach	High yield of adequate tissue Hemostasis is more secure	Requires general/spinal anesthesia; longer recovery period
Laparoscopic approach	High yield of adequate tissue Hemostasis is more secure	Requires general/spinal anesthesia; longer recovery period

A full assessment of the renal biopsy requires examination by light microscopy, immunohistochemistry, and electron microscopy (EM), with the use of other tests in some circumstances. Therefore it is necessary for the biopsy to be divided to provide material for each of these methods of examination. During this process it is extremely important that the biopsy is not damaged by handling, or by drying, and that the tissue is fixed in an appropriate fixative as quickly as possible, ideally within minutes. This is best achieved by dividing the biopsy at the bedside. Examination of the biopsy with a dissecting microscope allows cortex, containing glomeruli, to be distinguished from medulla and thus facilitates assessment of the adequacy of the cores and division of the biopsy so that glomeruli are present in the samples for each modality of examination. If a dissecting microscope is not available, then a standard light microscope can be used with the biopsy placed in a drop of normal saline on a microscope slide. If it is not possible to examine the biopsy in this way, then a standard approach to obtain material for EM is to take small fragments (approximately 1 mm in length) from each end of each core. In that way if there is cortex in the core, glomeruli should be sampled. The remainder of the cores can then be divided for light microscopy and for immunofluorescence (IF). The part of the biopsy for light microscopy is then placed in appropriate fixative and that for IF is either snap frozen or transported to the laboratory in a suitable transport medium such as that described by Michel et al.²⁷; tissue placed in this medium can remain for several days at room temperature without loss of antigens. During division of the biopsy it is important not to introduce artifacts due to crushing or stretching. Forceps should not be used to pick up the specimen; this can be done using either a needle or a small wooden stick such as a toothpick. The biopsy should be cut using a fresh scalpel.

If the biopsy has to be taken to the histology laboratory for division, this should be done as quickly as possible with the biopsy wrapped in saline-moistened gauze or in tissue

culture medium. Artifacts may be produced if the biopsy is placed on dry gauze or gauze moistened with water, or if it is placed in ice-cold saline.

If the amount of material obtained at biopsy is limited, then it may be necessary to adapt the way in which it is divided and the decision as to how this is done must depend on the clinical question. In most cases it is possible to omit frozen material for IF and instead perform immunohistochemistry on paraffin sections. However, if there is a suspicion of crescentic glomerulonephritis due to antiglomerular basement membrane (anti-GBM) disease, IF is more reliable for detecting the linear capillary wall staining. It may be possible to omit EM and perform it if necessary on material reprocessed from the paraffin block, but if this is done, it is not possible to obtain accurate measurements of glomerular capillary membrane thickness.²⁸

LIGHT MICROSCOPY

The most commonly used fixative for light microscopy is buffered 10% aqueous formaldehyde solution. This is actually a 10% solution of the 37% commercially available concentrated solution of formaldehyde, giving a final concentration of about 4%. This fixative is generally available in all histology laboratories, provides adequate fixation for light microscopy, and also allows the tissue to be used for immunohistochemistry and EM. Some more specialized fixatives such as Bouin or Zenker fixative provide better preservation of certain morphologic details, but in general the problems with handling these fixatives, and the difficulties of subsequently using the material for immunohistochemistry or EM, outweigh the advantages. For example, Bouin contains picric acid that is explosive when dry. However, we do commonly use Bouin fixative for examination of mouse kidneys where the improvement in glomerular morphology is significant. Methacarn, a modified Carnoy fixative, also provides good fixation for light microscopy and EM and may allow the immunohistochemical detection of antigens that are not detected in formalin-fixed tissue. Details of the preparation of various fixatives can be found in the appendix of Churg and colleagues.²⁴

The standard method of processing tissue for light microscopy is by dehydration in graded alcohols, transfer to a clearing agent such as xylene, and embedding in paraffin wax. This is usually performed in an automated instrument but can be done by hand. Rapid processing schedules allow for same-day processing and it is possible to obtain stained slides within 3 to 4 hours of receipt of the specimen in the laboratory.

It is important to have thin uniform sections for light microscopy. These should be cut as thin as possible – no greater than 3 μ m. It is often stated that renal biopsy sections should be cut at 2 μ m, but this may lead to problems in cutting with damage to the tissue. Since many pathologic lesions may be focal within glomeruli, interstitium, or vessels, it is essential that the biopsy is examined at multiple levels and each laboratory will have their preferred way to achieve that. In general, serial sections should be cut with at least two placed on each slide. Multiple slides can then be stained with each stain, with some intervening unstained sections kept either for potential immunohistochemical examination or for other special stains as necessary.

STAINING FOR LIGHT MICROSCOPY

Most renal pathologists employ a number of stains for light microscopy. The commonly used stains are hematoxylin and eosin (H&E), periodic acid–Schiff (PAS) reaction, silver methenamine, and a trichrome stain. The H&E stain is a good general histological stain for studying the overall architecture of the kidney. It is good for studying the morphology of tubular cells and the morphology of interstitial infiltrates. With experience the different staining characteristics of hyaline, fibrin, and amyloid, all of which are eosinophilic, can usually be distinguished. However, the H&E stain does not distinguish staining of glomerular matrix and basement membrane from cell cytoplasm, and therefore is less useful for the assessment of glomerular architecture. In the PAS reaction the mesangial matrix and basement membrane are stained purple and this allows a good assessment of the amount of matrix and the thickness of the GBM. PAS also stains the tubular basement membranes and hyaline deposits. The silver methenamine stain is the best stain for studying the detailed morphology of the GBM and for highlighting the membrane spikes seen in membranous glomerulonephritis and the double contours seen in membranoproliferative glomerulonephritis. Its only drawback is that a satisfactory result is more technically demanding than the other stains. A trichrome stain, such as Masson trichrome, will stain the glomerular mesangial matrix and basement membrane and may also help in highlighting fibrin and immune complex deposits. Other stains are a matter of personal preference. We always use an elastin stain to demonstrate the elastic laminae of vessels and this is counterstained with picrosirius red to stain fibrillar collagen in the interstitium. Amyloid is most specifically detected in a Congo red stain and we feel it is prudent to perform this in all native biopsies. This is the exception to the requirement for thin sections; since the Congo red stain is relatively insensitive, a section cut at 10 μm should be used. Details of staining methods are given in the appendix of Churg and colleagues.²⁴ Other stains that may be employed when necessary include the von Kossa stain that demonstrates calcium deposition and the Perls Prussian blue stain for iron.

EXAMINATION OF THE BIOPSY BY LIGHT MICROSCOPY

It is important to approach the examination of the biopsy systematically. Sections should first be assessed at low power to determine what parts of the kidney (or other structures in some cases) they contain, including whether there is cortex and/or medulla. A low-power view will also allow an assessment of the amount of chronic nephron damage, as demonstrated by tubular atrophy and interstitial fibrosis, and the presence of interstitial inflammatory infiltrates. It will also allow an assessment of interstitial expansion, most commonly due to either edema or fibrosis, but occasionally due to infiltration by, for example, amyloid. Examination should then proceed by studying the glomeruli, tubules, interstitium, and vessels, including arteries, arterioles, and veins, in more detail. Features that should be looked for in glomeruli and tubules are detailed in [Boxes 26.2](#) and [26.3](#). Arterioles should

Box 26.2 Features to Be Assessed by Light Microscopy in Glomeruli

- Size.
- Cellularity: if increased, then are the extra cells mesangial, in capillary lumens (endocapillary), or in Bowman's space? (NB: Normal mesangial areas contain two to three cells.)
- Capillary wall thickness (use periodic acid–Schiff or silver stain); if thickened, are there double contours or spikes on the silver stain?
- Is the mesangium expanded? If so, are there nodules?
- Is there deposition of abnormal material (e.g., amyloid)?
- Is there segmental sclerosis?
- Is there thrombosis?
- Is there necrosis?

Box 26.3 Features to Be Assessed by Light Microscopy in Tubules

- Percentage atrophy
- Signs of acute damage (e.g., dilatation, epithelial flattening, granular casts, mitoses)
- Tubulitis
- Casts: Granular casts suggest acute tubular injury; eosinophilic fractured casts suggest myeloma; neutrophil casts suggest acute pyelonephritis
- Crystals (e.g., oxalate)
- Viral inclusions (e.g., BK virus)

be examined for the presence of hyalinosis, thrombosis, and necrosis. Arteries should be assessed for intimal thickening and whether it is accompanied by reduplication of the internal elastic lamina, thrombosis, necrosis, inflammation, and cholesterol emboli.

TERMINOLOGY IN DESCRIPTION OF GLOMERULAR DISEASE

The involvement of glomeruli by a pathological process can be defined by the percentage of glomeruli involved by a lesion and by whether the lesion involves all or only part of any individual glomerulus. A lesion that involves all or nearly all glomeruli is described as “diffuse,” whereas one that involves some but not all glomeruli is described as “focal.” In the definitions given in the World Health Organization (WHO) atlas of glomerular diseases, it was suggested that the cutoff for focal versus diffuse should be 80% of glomerular involvement. However, in recent classifications of lupus glomerulonephritis²⁹ and immunoglobulin A (IgA) nephropathy,³⁰ the cutoff is defined as 50%. If a lesion involves only part of a glomerulus, that is, with some capillary lumens remaining uninvolved, it is called “segmental,” whereas if it involves the whole glomerulus, it is called “global.” In the classifications of lupus glomerulonephritis²⁹ and IgA nephropathy,³⁰ the cutoff is set at 50% glomerular tuft involvement, except for segmental sclerosis in IgA nephropathy, in which any area of sclerosis that leaves some of the glomerulus unaffected is defined as segmental.

Box 26.4 Definitions of Terms Used in Describing Glomerular Lesions

Sclerosis: A lesion resulting from an increase in mesangial matrix and/or collapse and condensation of the basement membranes – the sclerotic material stains with eosin, periodic acid–Schiff (PAS), and silver stains.

Hyalinosis: Lesion containing an acellular structureless material consisting of glycoproteins and sometimes lipids – stains intensely with eosin and PAS but not with silver stains.

Fibrosis: A lesion consisting of collagen fibers, which may be differentiated from sclerosis by not staining with PAS reagent or silver stains.

Necrosis: A lesion characterized by fragmentation of nuclei and/or disruption of the basement membrane, often associated with the presence of fibrin-rich material.

Extracapillary proliferation or cellular crescent: Extracapillary cell proliferation of more than two cell layers with >50% of the lesion occupied by cells.

Extracapillary fibrocellular proliferation or fibrocellular crescent: An extracapillary lesion comprising cells and extracellular matrix, with <50% cells and <90% matrix.

Extracapillary fibrosis or fibrous crescent: >10% of the circumference of Bowman's capsule covered by a lesion composed of >90% matrix.

There are a number of other terms, such as sclerosis and hyalinosis, that have specific definitions in the glomerulus and these are listed in [Box 26.4](#).

IMMUNOHISTOCHEMISTRY

The understanding of renal pathology was transformed in the 1960s by the use of IF microscopy. This allowed the detection and localization of immunoglobulins and complement components in glomeruli, and the identification of new entities such as IgA nephropathy. It is mandatory to perform immunohistochemistry for a full assessment of glomerular pathology in biopsies for native kidneys. The use of immunohistochemistry in transplant biopsies is discussed further later. There are a number of diagnoses that cannot be made on renal biopsy without immunohistochemistry including IgA nephropathy, C3 glomerulopathy (GN), C1q nephropathy, anti-GBM disease, and light-chain deposition disease.

In native kidneys a minimum panel of immunohistochemical stains would include antibodies for IgA, IgG, IgM, C3c, and kappa and lambda light chains. Light-chain immunohistochemistry is very important if diagnoses such as light-chain deposition disease, monoclonal immunoglobulin deposition disease, or proliferative glomerulonephritis with monoclonal immunoglobulin deposits are not to be missed. Many pathologists would also add antibodies for C1q, C4c, and fibrinogen to this routine panel. In transplant kidney biopsies staining for C4d is invaluable in assessing the activation of the classical pathway of complement by antibody and hence in the diagnosis of antibody-mediated rejection. There are a number of other antigens whose detection may be useful in particular circumstances. These include:

1. Microorganisms, including BK virus, cytomegalovirus, and Epstein–Barr virus.

2. Amyloid proteins. Antibodies are available to AA amyloid and many of the rarer inherited forms of amyloid.
3. Alpha chains of type 4 collagen. In suspected hereditary nephropathy of Alport type, it may be helpful to stain for the alpha 3 and alpha 5 chains of type IV collagen.
4. IgG subclasses in cases of suspected monoclonal immunoglobulin deposition.
5. Myoglobin in suspected myoglobinuria.
6. Lymphocyte surface antigens particularly in cases of suspected lymphoid neoplasia.
7. Type III collagen in collagenofibrotic GN.
8. Fibronectin in fibronectin GN.

Immunohistochemistry is performed either on cryostat sections of a piece of snap-frozen tissue or on paraffin sections. Antigen detection on frozen sections is usually performed using an antibody labeled with a fluorochrome and this is then viewed using a fluorescence microscope – commonly referred to as immunofluorescence or IF. The use of fluorescent-labeled antibodies on frozen sections is technically straightforward and very sensitive because the antigens have not been altered by fixation. There are some drawbacks. First, it requires a separate piece of tissue to have been obtained at the time of biopsy. Second, the morphology of frozen sections is never as good as paraffin sections and so it may be more difficult to define the site of the antigen within the glomerulus. In addition, immunofluorescent sections will fade over time but if appropriately mounted and refrigerated in the dark, they will retain staining for weeks to months.

If paraffin sections are used, then some form of antigen retrieval is essential for most antigens because they become “masked” during fixation and processing. For the detection of immunoglobulins and complement the antigen retrieval that works best is some form of protease digestion. The length of time required for protease digestion is critically dependent on a number of factors such as the length of time the biopsy has been in fixative and the particular processing schedule used; some of these may be difficult to control. This variability of the antigen retrieval process is the major drawback of immunohistochemistry on paraffin sections and means that results are highly dependent on the skills of the technician performing the staining. After the antigen retrieval step, antigens are generally detected using a primary antibody followed by a detection system that leads to the deposition of a colored reaction product that is visible by light microscopy. Commonly this product is developed by a reaction that uses the enzyme horseradish peroxidase, and hence this method is often referred to as “immunoperoxidase” staining. However, it is also possible to use fluorescent antibody staining on paraffin sections after antigen retrieval.³¹

The major advantage of immunohistochemistry on paraffin sections is that it is not necessary to take a separate piece of tissue for frozen section. In addition, it is possible to specifically localize antigens and compare this with adjacent sections examined by light microscopy. However, it is technically demanding and also it is significantly less sensitive for some antigens. Direct comparison of the two methods showed that detection of IgG, IgA, and C3c by immunoperoxidase on protease-digested deparaffinized sections of formaldehyde-fixed tissue is, with few exceptions, equal to IF on frozen sections.³² Our experience is that it is extremely difficult to get satisfactory staining for light chains

using peroxidase techniques on paraffin sections (although fluorescence on paraffin sections may be more successful) and that detection of the linear capillary wall staining of anti-GBM antibodies is more difficult in paraffin sections. It may also be more difficult to detect very early deposits of membranous glomerulonephritis in paraffin as compared with frozen tissue.

In our experience most renal pathologists find IF on frozen sections the most satisfactory way to detect immunoglobulins and complement but, regardless of preference, there will always be cases in which no material is available for frozen section, or the material is inadequate, and so laboratories should also be competent to carry out immunohistochemistry on paraffin sections.

In reporting immunohistochemical staining for immunoglobulins and complement, it is important to describe the site of staining in the glomerulus (e.g., mesangial or capillary wall), its nature (whether linear, finely or coarsely granular), and its intensity. For estimation of intensity most pathologists rely on semiquantitative subjective scale from 0 to 3+, but formal quantitation by image analysis may be useful for research. In addition to the glomerulus, staining should also be assessed in the tubules, especially the tubular basement membrane, interstitium, and vessels.

ELECTRON MICROSCOPY

EM is invaluable for assessing structural changes in the glomerulus and for identifying immune complexes, which are seen as areas of electron density. Although the importance of EM has become much reduced in other areas of surgical pathology, because of the development of immunohistochemistry, it remains an invaluable technique for the examination of glomeruli in biopsies in native kidneys and, increasingly, for the determination of causes of dysfunction in transplant kidneys. The part of the renal biopsy on which EM is to be performed is usually placed in separate fixative, although entirely satisfactory results can be obtained using material fixed in formalin. Most laboratories prefer either ice-cold glutaraldehyde or paraformaldehyde. The material is then exposed to osmium tetroxide and processed into resin blocks. In order to select the areas to be studied, “semithin” 0.5- μ m sections are first screened by light microscopy to select areas of interest that can then be examined further on the electron microscope. An ultramicrotome is then used to obtain the very thin sections required for the electron microscope. A permanent record of the electron microscopic appearances is kept either as photographs or, increasingly, as digital images. As with light microscopy, examination of the biopsy by EM should be systematic with assessment of the glomerular capillary basement membrane and its thickness; the endothelium and whether there is thickening or loss of fenestrations; the capillary lumen and particularly whether there is narrowing by cells or other material; the podocytes looking particularly at the preservation of the foot processes and whether the cell bodies show any vacuolation or microvillous change. The presence of any electron-dense deposits – most commonly due to immune complex deposition – should be noted together with their distribution – mesangial, subendothelial, or subepithelial. EM may also demonstrate a number of other structures such as fibrils in amyloidosis

or fibrillary glomerulonephritis, tubules in immunotactoid GN, or the characteristic inclusion bodies of various storage diseases.

Although EM is most useful in the assessment of glomerular morphology, it may also be very helpful in demonstrating ultrastructural changes in other parts of the kidney. For example, it may help in demonstrating tubular basement membrane immune complexes, in elucidating the nature of tubular epithelial cell inclusions, and in examining the morphology of mitochondria in tubular epithelial cells, which may show abnormalities in inherited conditions or as a result of drugs.

There have been several studies that have assessed the utility of EM in the assessment of native kidney biopsies. Most studies suggest that EM provides useful information in about half of all native kidney biopsies and is essential for diagnosis in about 20%.³³ Because it is impossible to know which these are at the time of biopsy, it is prudent to always have material available for EM even if, in some cases, it is not processed further after light microscopy and immunohistochemistry. [Box 26.5](#) lists some conditions for which EM is essential for the diagnosis and others where it is helpful. Also listed are some conditions in which the diagnosis may be reached without EM but, even in these cases, it is important to remember that EM may allow a more detailed description of the morphology of these conditions or also reveal a totally unrelated pathology.

Box 26.5 Examples of the Use of Electron Microscopy in Diagnosis of Kidney Biopsies

Electron Microscopy (EM) Essential for Diagnosis

Thin basement membrane lesion
Fibrillary glomerulopathy (GN)
Immunotactoid GN
Alport syndrome
Fabry disease
Lecithin-cholesterol acetyltransferase deficiency
Nail patella syndrome

EM Very Helpful

Dense deposit disease
Minimal change disease
Early diabetic GN
Early membranous GN (particularly if only paraffin sections are available for immunohistochemistry)
Membranoproliferative GN
Postinfectious GN
Human immunodeficiency virus (HIV) nephropathy
Lipoprotein GN
Collagenofibrotic GN

Diagnosis May Be Made Without EM

IgA nephropathy
Acute tubulointerstitial nephritis
Myeloma cast nephropathy
Pauci-immune crescentic GN
Amyloid (although amyloid fibrils may be detected by EM when it has been missed on light microscopy)

Morphometric analysis of EM is mainly of importance for research. However, it is important to be able to measure the thickness of the GBM in order to quantitate the thinning that may be seen in thin basement membrane lesion or the thickening commonly seen in diabetic glomerulosclerosis. Accurate unbiased measurement of the GBM thickness requires complex morphometric techniques to avoid the bias introduced by tangential sectioning of capillary loops. However, in practice it is satisfactory to use direct measurement of GBM thickness (distance from endothelial to podocyte plasma membrane) and determination of the arithmetic mean of such measurements. Das et al.³⁴ found that if 16 measurements from each of two glomeruli were made using this direct method, the results were reproducible. Ideally, each laboratory should define a normal range using this method.

OTHER STUDIES ON THE RENAL BIOPSY

In addition to examination by light microscopy, immunohistochemistry, and EM it may also be appropriate to consider other methods for studying the tissue. In cases of suspected infection, part of the biopsy may be sent for culture or for polymerase chain reaction for infective organisms. In biopsies with lymphoid infiltrates, immunoglobulin gene rearrangement studies may allow the confirmation of clonality. The chemical composition of material in the biopsy (e.g., crystalline material) may be determined by energy-dispersive x-ray spectroscopy.

There has been considerable interest in the possibilities of extracting messenger RNA (mRNA) from biopsies, in order to study differences in gene expression in different pathological conditions,³⁵ and in studying the range of proteins in the biopsy – the proteome.³⁶ These techniques have been applied to whole biopsies or to parts of the biopsy – for example, glomeruli isolated either by simple dissection under a dissecting microscope or by laser capture microdissection. Examination by mass spectrometry may allow identification of the fibril proteins of amyloid or fibrillary glomerulonephritis.^{37,38} Examination of mRNA transcripts shows considerable promise for assisting in the diagnosis of kidney transplant rejection.³⁹

BIOPSIES OF TRANSPLANTED KIDNEYS

The handling of transplant biopsies differs in some respects from that of native kidneys. For biopsies taken to assess the cause of kidney dysfunction in the first few months after transplantation, it may not be necessary to carry out immunohistochemistry with a full panel of antibodies to immunoglobulins and complement, or to perform EM, unless there is a clinical suspicion of glomerular disease. However, immunohistochemistry for C4d is performed to assess antibody binding and complement activation on peritubular capillary endothelium. In later biopsies EM is very useful in the diagnosis of chronic allograft GN and its differentiation from recurrence of *de novo* glomerulonephritis. It is also helpful in identifying chronic rejection involving peritubular capillaries, which is associated with multilayering of the peritubular capillary basement membranes.⁴⁰ The recommendations from the Banff Conference on Allograft Pathology 2013⁴¹ are that ultrastructural studies should be performed in all biopsies

from patients who are sensitized, have documented donor-specific antibodies at any time after transplantation, and/or who have had a prior biopsy showing C4d staining, glomerulitis, and/or peritubular capillaritis. It is also advised that EM be considered in all biopsies performed more than 6 months after transplantation and in “for-cause” biopsies done more than 3 months after transplantation to determine if early changes of transplant GN are present, prompting testing for donor-specific antibodies.

THE SIZE OF THE BIOPSY

The renal biopsy is only a small sample of the renal parenchyma and this always needs to be kept in mind when making inferences about the state of the whole kidney from changes seen in the biopsy. Some diseases may only affect the kidney focally and therefore may be missed on biopsy, for example, reflux nephropathy or arterial cholesterol emboli. Others may be segmental at the level of the glomerulus, for example, focal and segmental glomerulosclerosis or pauci-immune necrotizing glomerulonephritis, and therefore the chance of detecting them will depend on how many glomeruli are present in the biopsy and how many sections are examined. Sampling is also a problem when we make inferences from the amount of disease we see in the biopsy to the amount that affects the kidney. For example, if 20% of the glomeruli in a biopsy have crescents, we tend to assume that this is the percentage of the glomeruli in the kidney that have crescents. However, because of the small size of most biopsies, the confidence limits we can place on the true involvement of glomeruli are usually very wide. An elegant mathematical description of the problems of glomerular sampling has been published by Corwin et al.⁴² This shows, for example, that to confidently exclude a segmental glomerular disease that is affecting about 5% of the glomeruli, a biopsy with 20 glomeruli is needed. The situation is worse if we consider the problem of comparing the amount of glomerular involvement in two different biopsies, a question that often arises, for example, in patients with lupus glomerulonephritis who have repeat biopsies. In that case, to confidently detect a 10% difference in glomerular involvement between two biopsies would require over 100 glomeruli in each biopsy. To detect differences of 25% to 40% glomerular involvement the minimum biopsy size is 20 to 25 glomeruli.

For some diseases, classification schemes have defined minimum sizes for biopsy adequacy. Thus, for lupus glomerulonephritis it is suggested that a biopsy should contain a minimum of 10 glomeruli.²⁹ In transplant biopsies the Banff group has suggested that the requirements for biopsy adequacy are 10 or more glomeruli with at least two arteries.⁴³ It has been shown that examining two rather than one core of tissue increases the sensitivity for the diagnosis of acute rejection from 91% to 99%.⁴⁴ In acute cellular rejection, examining slides taken at only one level, rather than at three, misses 33% of cases with intimal arteritis.⁴⁵

THE BIOPSY REPORT

The biopsy report should include a morphologic description of the biopsy and an interpretation of the appearances in

the light of the clinical presentation. The changes seen on light microscopy, immunohistochemistry, and EM must be integrated and this is best done if a single person examines the biopsy by each modality. A Committee of the Renal Pathology Society has published recommendations for the essential elements that should be present in a renal biopsy report.⁴⁶

The description of the light microscopy should include the number of glomeruli present and the number that show global or segmental sclerosis. It is essential to provide a quantitative estimate of the amount of irreversible nephron damage in the biopsy and, where appropriate, the severity of any active inflammatory process. The best way to estimate the irreversible damage is by specifying the number of globally sclerosed glomeruli and the amount of tubular atrophy and interstitial fibrosis. The estimate of activity will depend on the particular disease process but should include an indication of the proportion of glomeruli involved by crescents, necrosis, and endocapillary hypercellularity. For some diagnoses there are established classification schemes that should be applied to the biopsy, for example, the International Society of Nephrology (ISN)/Renal Pathology Society (RPS) classification of lupus nephritis,²⁹ the Oxford Classification of IgA nephropathy,^{47,48} and the Banff classification of allograft pathology. International consensus groups have published recommendations for Pathologic Classification, Diagnosis, and Reporting of GN⁴⁹ and for the standardized grading of chronic changes in native kidney biopsy specimens.⁵⁰

The interpretation of renal biopsies requires the pathologist to integrate the biopsy findings with detailed clinical information and therefore requires a thorough understanding of renal disease and the therapeutic implications of the biopsy diagnosis. Close communication between the clinician and pathologist is essential and it is generally very helpful for the biopsy to be viewed and discussed at a clinicopathologic conference so that full discussion of the implications of the

biopsy specimen appearances for patient management can take place.

CONCLUSION

Percutaneous renal biopsy is generally safe if care is taken to select and prepare the patients beforehand. It has become a cornerstone of nephrological practice and its handling and interpretation should be made by those experienced in renal pathology. The interpretation of the biopsy should be carried out with adequate clinical information for integrated clinicopathologic conclusions to be drawn.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. Relative contraindications to percutaneous renal biopsy include the following:

- a. Uncontrolled hypertension
- b. Bleeding diathesis
- c. Hydronephrosis
- d. Single kidney
- e. Uncooperative patient

Answer: d

Rationale: All the others are absolute contraindications (Table 26.2).

2. The best stain for viewing glomerular architecture using light microscopy is:

- a. Hematoxylin and eosin (H&E)
- b. Periodic acid–Schiff (PAS)
- c. Silver methenamine
- d. Masson trichrome
- e. Congo red

Answer: b

Rationale: PAS distinguishes the mesangial matrix and glomerular basement membrane from the other components of the glomerulus and therefore allows good assessment of amount of mesangial matrix, mesangial hypercellularity, and the thickness of the basement membrane.

3. Pathology that affects only part of a glomerulus is termed:

- a. Segmental
- b. Mild
- c. Limited

d. Focal

e. Sclerosis

Answer: a

Rationale: A lesion that affects only part of a glomerulus, that is, with some capillary lumens remaining uninvolved, is called segmental.

4. Electron microscopy is not essential for making a diagnosis of:

- a. Alport syndrome
- b. Thin membrane disease/lesion
- c. Immunoglobulin A nephropathy
- d. Immunotactoid glomerulopathy
- e. Fibrillary glomerulopathy

Answer: c

Rationale: See Box 26.5.

5. In a renal biopsy, a lesion containing an acellular structureless material consisting of glycoproteins and sometimes lipids that stains intensely with eosin and periodic acid–Schiff (PAS) but not with silver stains is termed:

- a. Sclerosis
- b. Fibrosis
- c. Necrosis
- d. Amyloid
- e. Hyalinosis

Answer: e

Rationale: See Box 26.4.